MTH309 PRACTICE PROBLEMS WEEK 11

Question 1. Let $X, X_1, X_2, \cdots$ be real valued random variables on the same probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that $X_n \xrightarrow[n\to\infty]{\mathbb{P}} X$. In addition, assume that there exists $Y \in \mathcal{L}^p(\mathbb{P})$ such that a.s. $|X_n| \leq Y$ for all n. Prove that $X \in \mathcal{L}^p(\mathbb{P})$. Is it true that $X_n \xrightarrow[n\to\infty]{\mathcal{L}^p(\mathbb{P})} X$?

Question 2. Given an RV X and $\alpha \geq 1$, prove the following statements.

(a)
$$\mathbb{E}|X|^\alpha = \int_0^\infty \mathbb{P}(|X|^\alpha > x)\, dx = \alpha \int_0^\infty x^{\alpha-1}\mathbb{P}(|X| > x)\, dx.$$

(b)
$$\mathbb{E}|X|^\alpha < \infty \Leftrightarrow \sum_{n=0}^\infty \mathbb{P}(|X|^\alpha > n) < \infty \Leftrightarrow \sum_{n=0}^\infty \mathbb{P}(|X| > n^{\frac{1}{\alpha}}) < \infty.$$

(c)
$$\mathbb{E}|X|^\alpha < \infty \Rightarrow \lim_{n\to\infty} n^\alpha \mathbb{P}(|X| > n) = 0.$$

Question 3. Let X be an RV with distribution function F_X. Fix $c \in \mathbb{R}$. Compute
$$\int_{\mathbb{R}} [F_X(x+c) - F_X(x)]\, d\lambda(x),$$
where λ denotes the Lebesgue measure on $\mathbb{R}$.

Question 4. Find examples of RVs X and Y, defined on the same probability space, which are uncorrelated but not independent.

Question 5. Let X and Y be independent random variables defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Further assume that $X, Y \in \mathcal{L}^2(\mathbb{P})$. Show that $Var(X+Y) = Var(X) + Var(Y)$.

Question 6. Let X and Y be two real valued random variables defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Prove the following two statements.

(a) X and Y are independent if and only if $\sigma(X)$ and $\sigma(Y)$ are independent. Here,
$$\sigma(X) = \{X^{-1}(B) : B \in B_{\mathbb{R}}\}.$$

(b) Given any measurable functions $f, g : \mathbb{R} \to \mathbb{R}$, independence of X and Y implies the independence of $f(X)$ and $g(Y)$.